Experiment 7 - Using Ansible

1. Update Your System

Open your terminal and run:

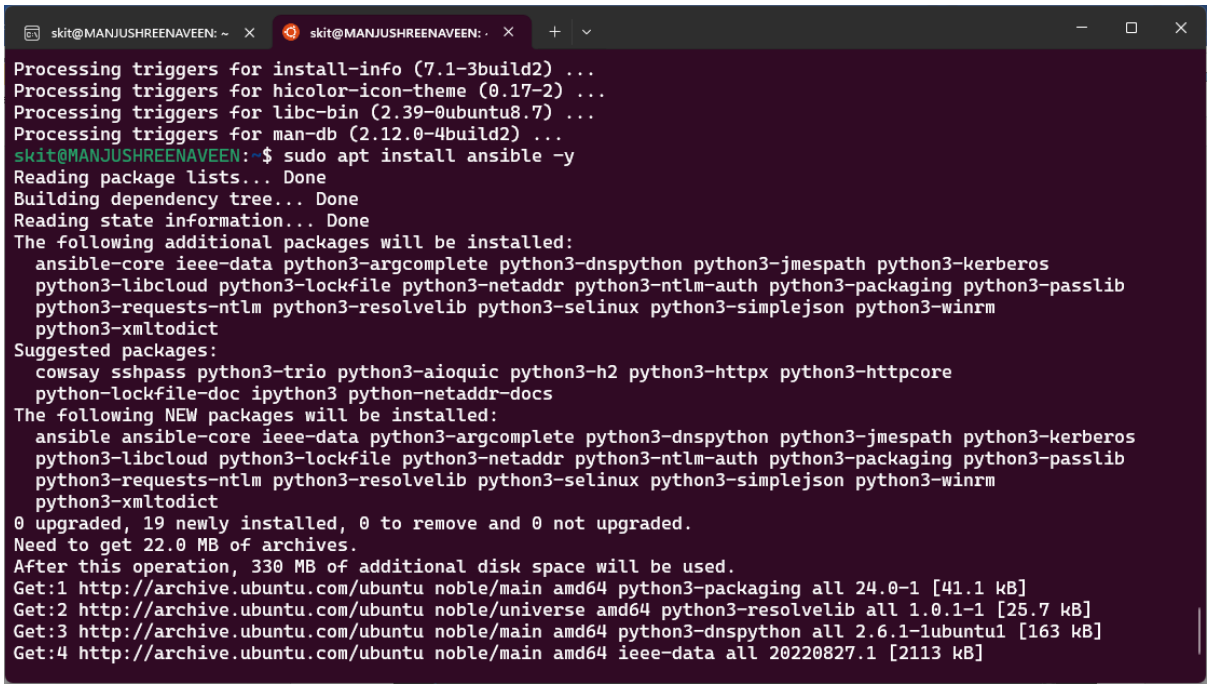
```
sudo apt update
sudo apt upgrade -y
```

---

2. Install Ansible

Install Ansible using apt:

```
sudo apt install ansible -y
```



Verify the installation by checking the version:

```
ansible --version
```

Expected Output Example:

```
Setting up python3-kerberos (1.1.14-3.1build9) ...
Setting up python3-simplejson (3.19.2-1build2) ...
Setting up python3-xmltodict (0.13.0-1ubuntu0.24.04.1) ...
Setting up python3-packaging (24.0-1) ...
Setting up python3-jmespath (1.0.1-1) ...
Setting up ieee-data (20220827.1) ...
Setting up python3-dnspython (2.6.1-1ubuntu1) ...
Setting up python3-selinux (3.5-2ubuntu2.1) ...
Setting up python3-argcomplete (3.1.4-1ubuntu0.1) ...
Setting up python3-requests-ntlm (1.1.0-3) ...
Setting up python3-libcloud (3.4.1-5) ...
Setting up python3-netaddr (0.8.0-2ubuntu1) ...
Setting up python3-winrm (0.4.3-2) ...
Setting up ansible-core (2.16.3-0ubuntu2) ...
Setting up ansible (9.2.0+dfsg-0ubuntu5) ...
Processing triggers for man-db (2.12.0-4build2) ...
skit@MANJUSHREENAVEEN:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/skit/.ansible/plugins/modules', '/usr/share/ansible/plugins/module
s']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/skit/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Mar  3 2026, 12:15:18) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
skit@MANJUSHREENAVEEN:~$
```

---

3. Creating an Ansible Inventory

An inventory file lists the hosts you want to manage. For this experiment, you can use the local host.

Create an Inventory File

1. Open your text editor to create a file called `hosts.ini`:

```
gedit hosts.ini
```

1. Add the following content to define the local host:

```
[local]
localhost ansible_connection=local
```

Explanation:

- `[local]` is a group name.
- `localhost` is the target host.
- `ansible_connection=local` tells Ansible to execute commands on the local machine without SSH.

3. Save and exit the file.

---

4. Writing a Basic Ansible Playbook

You will now create a simple playbook that performs two common tasks:

- Updating the apt cache
- Installing a package (e.g., curl)

---

5. Create the Playbook File

1. Open your text editor to create a file called `setup.yml`:

```
gedit setup.yml
```

1. Add the following YAML content:

```
---
- name: Basic Server Setup
  hosts: local
  become: yes # Use privilege escalation (sudo)
  tasks:
    - name: Update apt cache
      apt:
        update_cache: yes

    - name: Install curl
      apt:
        name: curl
        state: present
```

Explanation:

- `name`: Provides a descriptive name for the play.
- `hosts`: Specifies the group or hosts from the inventory file.
- `become: yes`: Uses sudo to perform tasks that require elevated privileges.
- Tasks Section:
  - Update apt cache: Uses the `apt` module to update the package cache.
  - Install curl: Uses the `apt` module to install the `curl` package if it isn't already installed.

1. Save and exit the file.

---

6. Running the Ansible Playbook

1. Execute the Playbook

In your terminal, run the following command:

```
sudo ansible-playbook -i hosts.ini setup.yml
```

Explanation:

- `ansible-playbook`: The command to run an Ansible playbook.
- `-i hosts.ini`: Specifies the inventory file.
- `setup.yml`: The playbook file you just created.

Expected Output:

An output that details each task. For example, a task summary might show "ok=2 changed=1" (if the apt cache was updated and curl was installed).

```
skit@MANJUSHREENAVEEN: ~    skit@MANJUSHREENAVEEN: ~

  configured module search path = ['/home/skit/.ansible/plugins/modules', '/usr/share/ansible/plugins/module
s']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/skit/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Mar  3 2026, 12:15:18) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
skit@MANJUSHREENAVEEN:~$ gedit hosts.ini
skit@MANJUSHREENAVEEN:~$ gedit setup.yml
skit@MANJUSHREENAVEEN:~$ sudo ansible-playbook -i hosts.ini setup.yml

PLAY [Basic Server Setup] ******************************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [localhost]

TASK [Update apt cache] ********************************************************************************
ok: [localhost]

TASK [Install curl] ************************************************************************************
ok: [localhost]

PLAY RECAP *********************************************************************************************
localhost                  : ok=3    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ign
ored=0

skit@MANJUSHREENAVEEN:~$
```